STA301 Formulae For Mid_Term

Date:

Cumulative frequency is obtained by adding previous to current frequency.

For Example:

X	f	CF
1	6	6
2	7	13
3	8	21
4	9	30
5	10	39

Prepared By

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Formula of mean :

$$\mu = \frac{\Sigma f x}{\Sigma f}$$

Range $= \text{Largest value} - \text{Smallest value}$

$= X_m - X_0$

Class Interval :

Denoted by h.

$$h = \frac{\text{Range}}{\text{No. of classes}}$$

* **Lower class**: It is the smallest value.

* **Upper class**: It is the value higher than the smallest value

* **Class Boundary**: It is obtained by subtracting 0.5 from lower_class and adding 0.5 to upper_class

* **Relative Frequency**:

$$\frac{\text{frequency}}{\text{sum of all frequencies}}$$

* For histogram we will use class boundaries and for frequency polygon we'll use mid points.

Midpoint: $\dfrac{\text{upper class limit} + \text{Lower class limit}}{2}$

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* Mode ($\hat{X}$) in case of raw data :

It will contain the most repeated value.

* Mode ($\hat{X}$) in case of frequency distribution of a continuous variable

$$\hat{X} = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$$

where :

l = lower class boundary

f_1 = frequency before model class

f_m = max frequency

f_2 = frequency after model class

h = class interval

* Arithmetic Mean ($\bar{X}$) :

$$\bar{X} = \frac{\Sigma x}{n}$$

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* Arithmetic Mean in case of frequency distribution:

$$\bar{X} = \frac{\Sigma f x}{\Sigma f}$$

* <u>Mid point / class Mark:</u>

$$x = \frac{\text{upper class} + \text{lower class}}{2}$$

* <u>Weighted Mean</u> :

$$\bar{X}_w = \frac{\Sigma w x}{\Sigma w}$$

* <u>Median $(\tilde{X})$</u>: (For odd values)

Median is the middle value of series.

<u>Median in case of Even values</u>:

Add both middle values and divide by 2.

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Median for Grouped Data:

$$\tilde{x} = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$

Consider:

Mileage Rating	f	Class Boundary	C.f
30.0–32.9	3	29.95–32.95	2
33.0–35.9	4	32.95–35.95	6
36.0–38.9	14	35.95–38.95	20
39.0–41.9	8	38.95–41.95	28
42.0–44.9	2	41.95–44.95	30

To find class boundary:

$= 33.0 - 32.9/2$
$= 0.1/2$
$= 0.05$

Subtract 0.05 from 30.0 and add 0.05 to 32.9.

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Where
$n = \Sigma f$
$n = 30$
$\frac{n}{2} = \frac{30}{2}$
$\boxed{\frac{n}{2} = 15}$

frequency table ma 15 k nearest value 14 ha.

f	Class-Boundary	C.F
4	32.95 - 35.95	6 c
f 14	l 35.95 - 38.95	20
8	38.95 - 41.95	28

$\tilde{x} = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$

* l (lower class boundary) = 35.95

* h [Class interval] = 38.95 - 35.95
= 3

* f (frequency) = 14

* c = 6

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First Quartile :

$$Q_1 = l + \frac{h}{f}\left(\frac{n}{4} - c\right)$$

Second Quartile :

$$Q_2 = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$

which is also the formula of median.

Third Quartile :

$$Q_3 = l + \frac{h}{f}\left(\frac{3n}{4} - c\right)$$

Deciles :

$$D_1 = l + \frac{h}{f}\left(\frac{n}{10} - c\right)$$

$$D_2 = l + \frac{h}{f}\left(\frac{2n}{10} - c\right)$$

$$D_3 = l + \frac{h}{f}\left(\frac{3n}{10} - c\right)$$

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Percentiles :

$$P_1 = l + \frac{h}{f}\left(\frac{n}{100} - c\right)$$

$$P_1 = l + \frac{h}{f}\left(\frac{2n}{100} - c\right)$$

$$P_3 = l + \frac{h}{f}\left(\frac{3n}{100} - c\right)$$

Geometric Mean for ungroup data:

Simple formula :

$$G = \sqrt[n]{x_1 \cdot x_2 \cdots\cdots x_n}$$

Logarithm formula :

$$G.M = \text{Antilog}\left[\frac{\Sigma \log x}{n}\right]$$

Date:

Geometric Mean for Grouped Data:

Note :

* Ungroup data aisa data hota ha jismay frequency given nahi hoti.

* Grouped data aisa data hota ha jismay frequency given hoti ha.

$$G.M = \text{Antilog}\left[\frac{\Sigma f \log x}{\Sigma f}\right]$$

* To find percentage average we will apply formula of geometric mean.

* To find speed/ average mean we will use the formula of harmonic mean.

* Harmonic Mean :

* For Un-group data :

$$H.M = \frac{n}{\Sigma\left(\frac{1}{x}\right)}$$

* For Grouped data :

$$H.M = \frac{\Sigma f}{\Sigma f\left(\frac{1}{x}\right)}$$

* Relation between Arithmetic mean ($\bar{x}$), G.M and H.M :

$$A.M(\bar{x}) \geq G.M \geq H.M$$

* Mid range $= \frac{X_m + X_o}{2}$

where X_m is the largest value and X_o is the smallest value.

Mid Quartile Range :

$$= \frac{Q_3 + Q_1}{2}$$

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* **Coefficient of dispersion:**

$$= \frac{X_m - X_0}{X_m + X_0}$$

* **Quartile Deviation:**

$$Q.D = \frac{Q_3 - Q_1}{2}$$

Also called semi interquartile range.

* **Coefficient of Quartile Deviation:**

$$= \frac{Q_3 - Q_1}{Q_3 + Q_2}$$

* **Mean Deviation:** (For ungrouped data)

$$M.D = \frac{\Sigma |d|}{n} \quad (d = x - \bar{x})$$

* **Mean Deviation for grouped data:**

$$M.D = \frac{\Sigma f |d|}{n}$$

Date:

* Coefficient of Mean Deviation:

$$= \frac{\text{Mean deviation}}{\text{Mean}}$$

* Variance:

$$= \frac{\sum (x - \bar{x})^2}{n}$$

* Standard Deviation: (For ungroup)

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

* Standard Deviation for grouped data:

$$S = \sqrt{\frac{\sum f (x - \bar{x})^2}{n}}$$

$$\Rightarrow \sqrt{\frac{\sum f x^2}{n} - \left(\frac{\sum f(\bar{x})}{n}\right)^2}$$

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* **Coefficient of S.D:**

$$= \frac{S.D}{mean}$$

* **Coefficient of Variance:**

$$C.V = \frac{S}{\bar{X}} \times 100$$

* **Pearson's Coefficient of skewness:**

Before applying empirical relation.

$$= \frac{mean - mode}{S.D}$$

* **In case if mode is not given:**

$$= \frac{3(mean - median)}{S.D}$$

→ After applying empirical relation.

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Bowley's Coefficient of skewness:

$$= \frac{Q_1 + Q_3 - 2\tilde{X}}{Q_3 - Q_1}$$

Percentile Coefficient of Kurtosis:

$$K = \frac{Q.D}{P_{90} - P_{10}}$$

Moments:

$$m_1 = \frac{\Sigma (x_i - \bar{x})}{n}$$

$$\vdots$$

$$m_r = \frac{1}{n} \Sigma (x_i - \bar{x})^r$$

Equation of Straight line:

$$y = a + bx$$

Date:

Standard Error of Estimate:

$$S_{y.x} = \sqrt{\frac{\Sigma (Y-\hat{Y})^2}{n-2}}$$

$$S_{y.x} = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

Pearson Coefficient of correlation:

$$r = \frac{\Sigma XY - (\Sigma X)(\Sigma Y)/n}{\sqrt{[\Sigma x^2 - (\Sigma x)^2/n][\Sigma Y^2 - (\Sigma Y)^2/n]}}$$

Range of r : $-1 < r < 1$

Positive Correlation : $0 < r < 1$

NO Correlation : $r = 0$

Negative Correlation : $-1 < r < 0$

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Law of Addition :

$$P(A \cup B) = P(A) + P(B)$$

Conditional Probability :

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Multiplication Law :

$$P(A \cap B) = P(A)\, P(B/A)$$

$$P(A \cap B) = P(B)\, P(A/B)$$

Multiplication Theorem of Probability:

$$P(A \cap B) = P(A)\,P(B)$$

Bayes' Theorem:

$$P(A_1|B) = \frac{P(A_1)\cdot P(B|A_1)}{P(A_1)\cdot P(B|A_1) + P(A_2)\cdot P(B|A_2)}$$

Properties of descrete probability distribution:

* $0 \leq P(X_i) \leq 1$ (i can be any number)

* $\Sigma P(X_i) = 1$

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Standard Deviation in case of probability distribution :

$$\sigma = S.D.(X) = \sqrt{\frac{\sum X^2 P(X)}{\sum P(X)} - \left[\frac{\sum X P(X)}{\sum P(X)}\right]^2}$$

$$\text{OR} \quad = \sqrt{\sum X^2 P(X) - \left[\sum X P(X)\right]^2}$$

Note : where "$\sum P(X) = 1$"